SSM Optimization

Weekly Meeting

17/04/26

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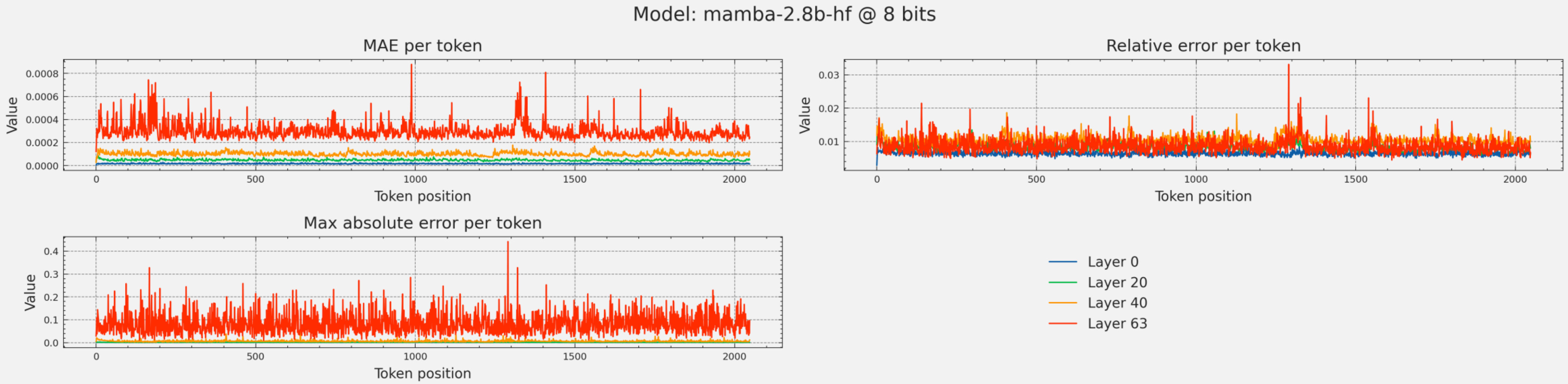
Experiment setup

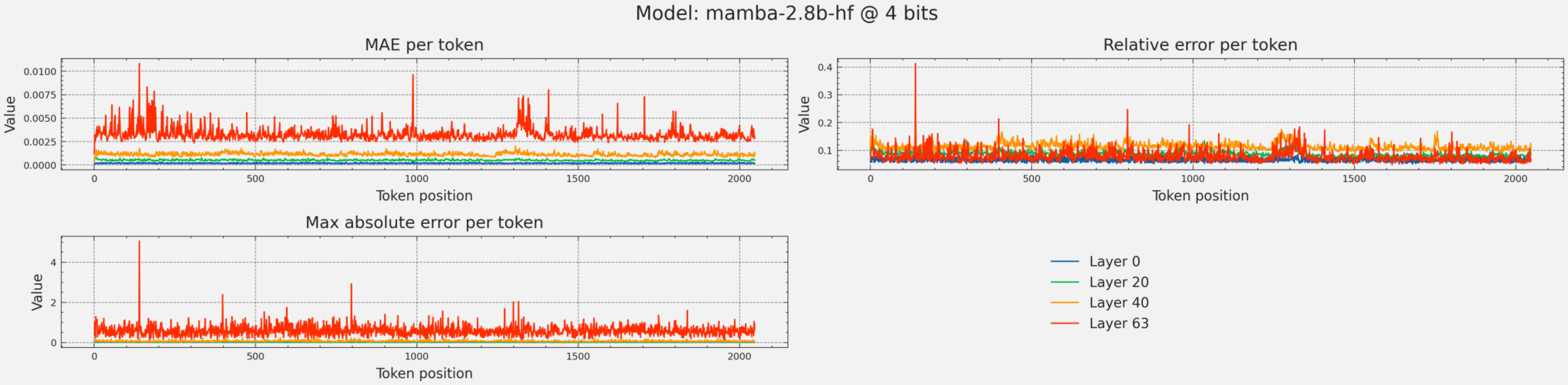
- Run inference over a fixed sequence and examine the state divergence of few SSM blocks of the network
- Checked on some Wikitext2 batches, and custom prompts
- Considered models: Mamba-2.8b and Falcon-Mamba-7b

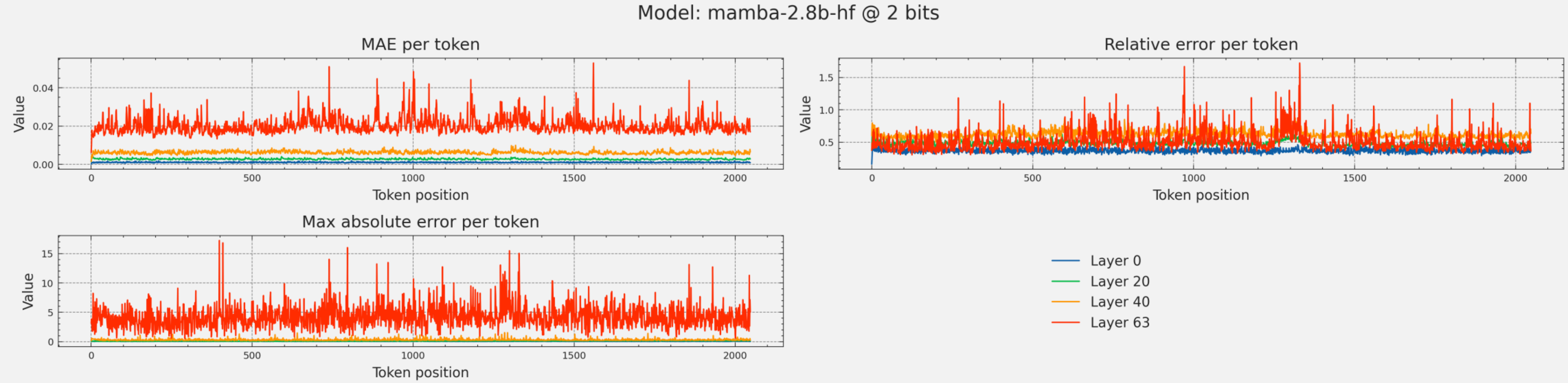
Mamba-2.8 / Seq. 2048 from Wikitext2

- *Input*: Wikitext2 - one batch of sequence length 2048
- *Model*: mamba-2.8b-hf
  Considered the state matrix of the SSM block of 4 layers



State drift

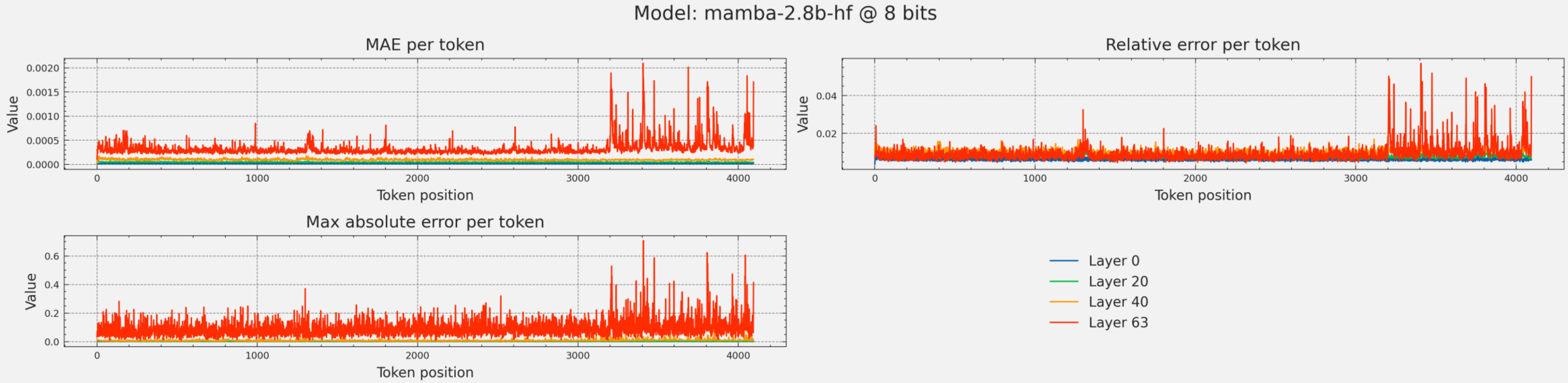




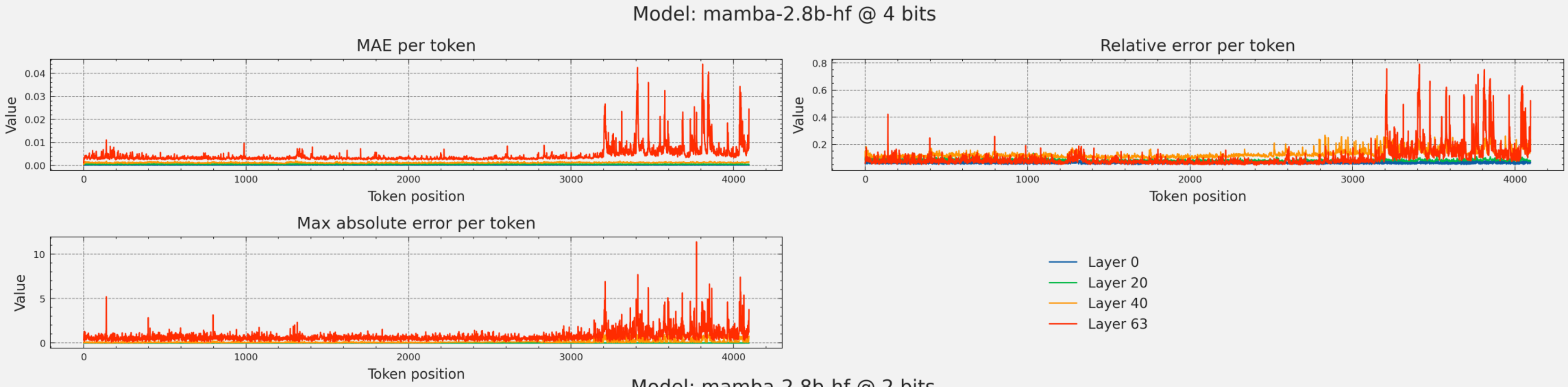
No particular drift for bit-widths in {2,4,8}

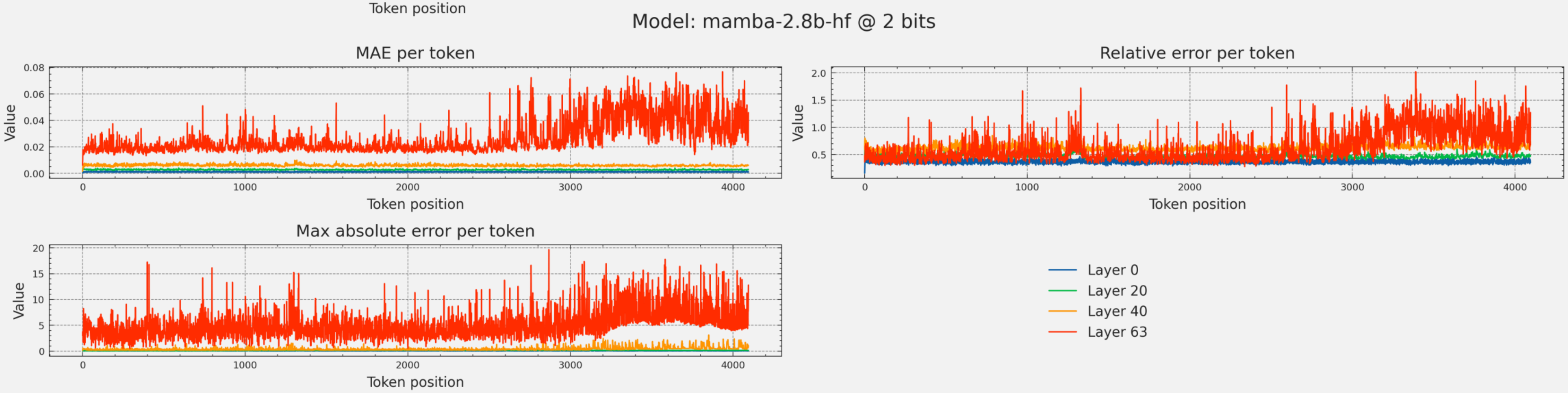
Mamba-2.8 / Seq. 4096 from Wikitext2

- *Input*: Wikitext2 - one batch of sequence length 4096
- *Model*: mamba-2.8b-hf
  Considered the state matrix of the SSM block of 4 layers



State drift

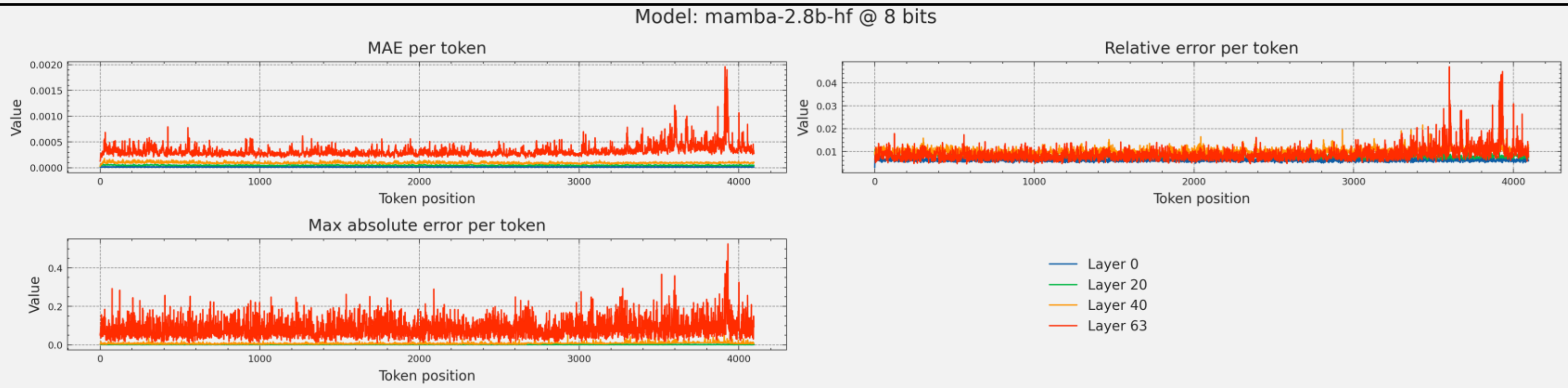


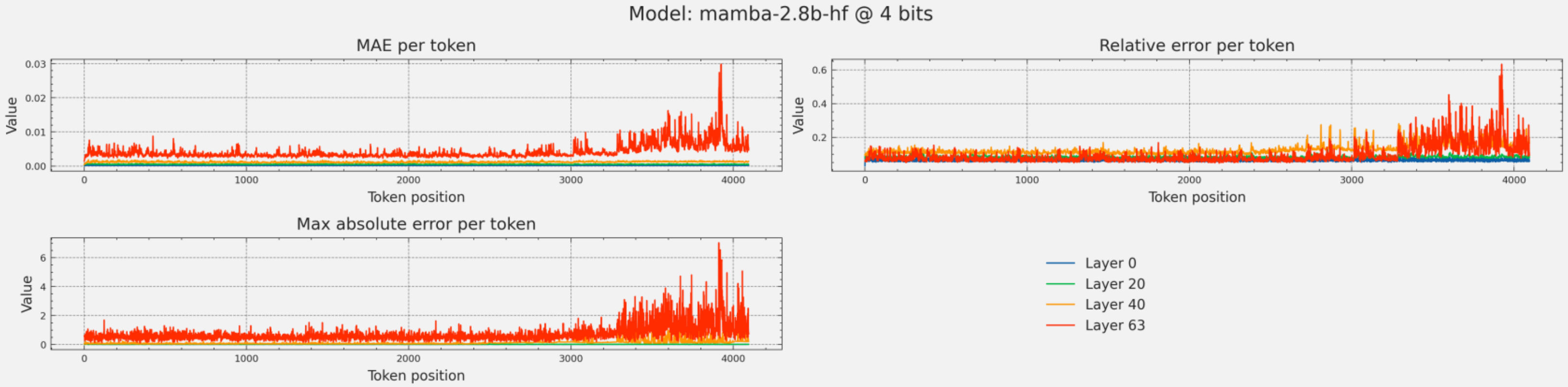


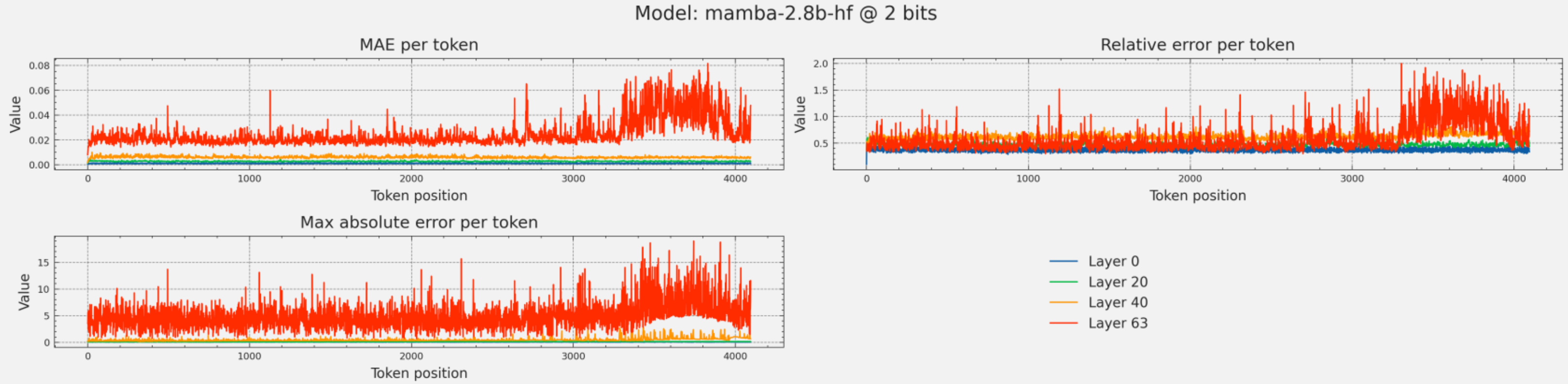
Starting from different token positions, at all bit-widths drift appears for tokens > #2500

State drift

Another batch

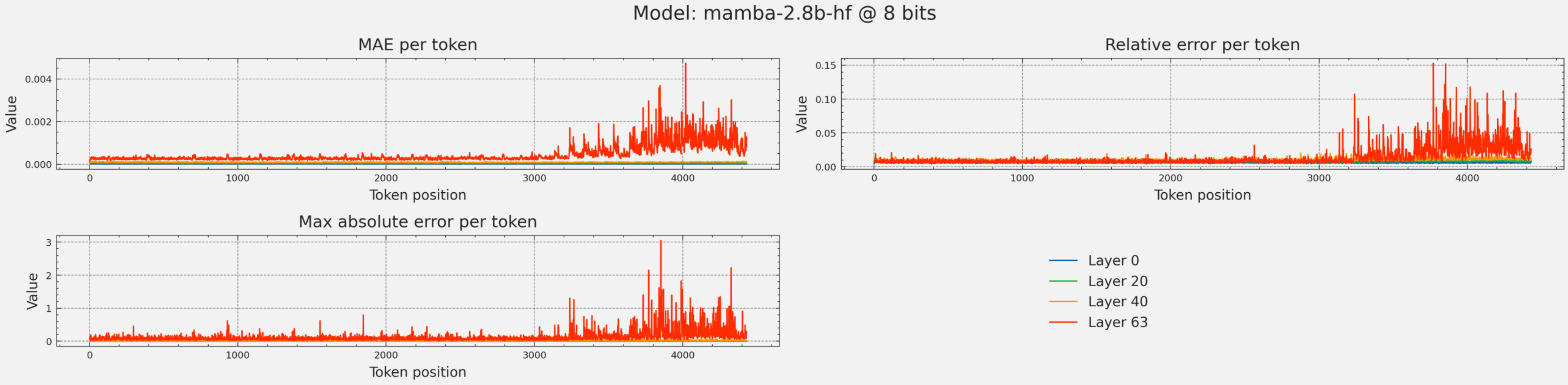




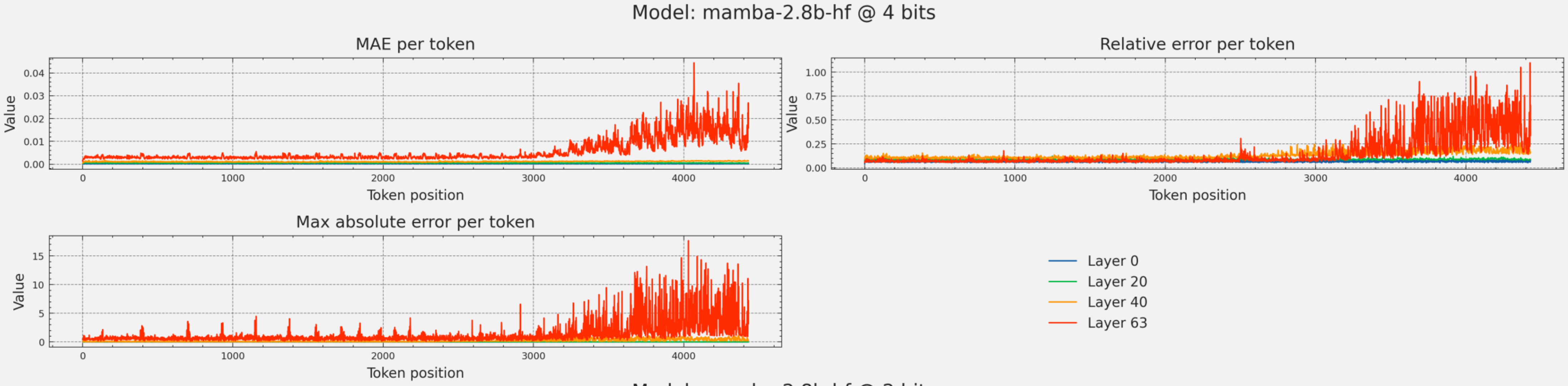


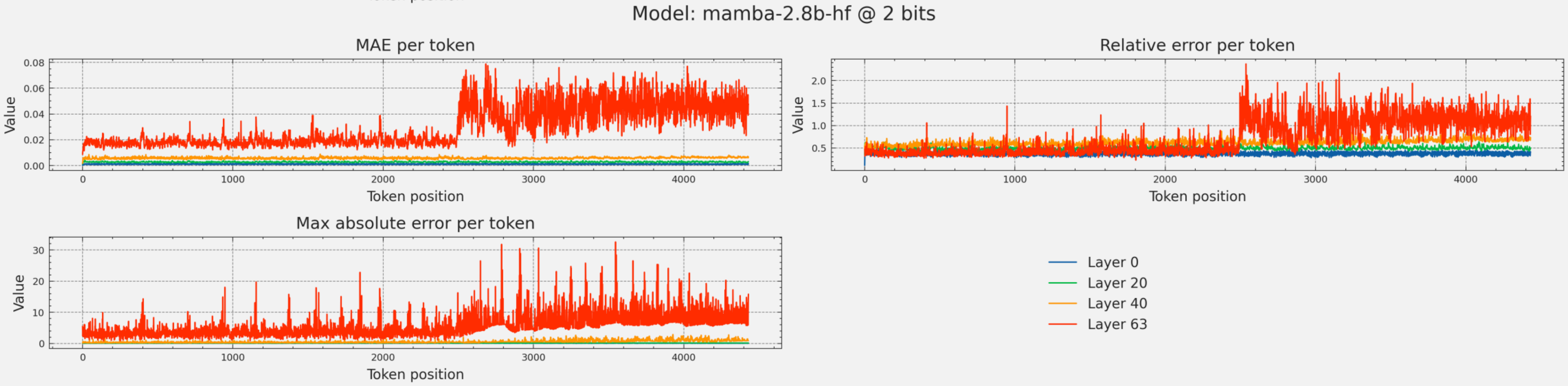
Mamba-2.8 / Seq. 4431 custom

- *Input*: custom prompt for long-range dependency
- *Model*: mamba-2.8b-hf
  Considered the state matrix of the SSM block of 4 layers



State drift

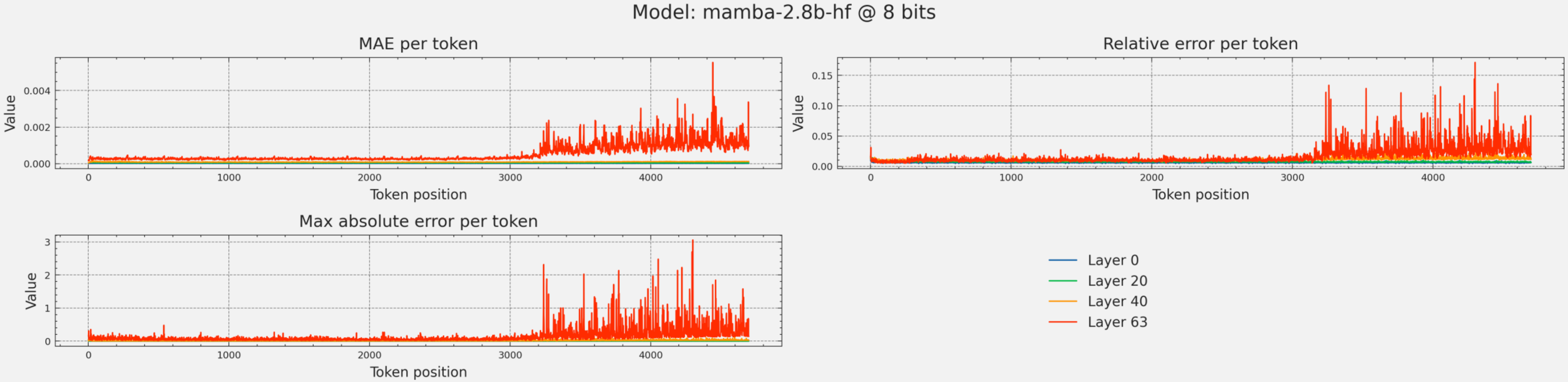




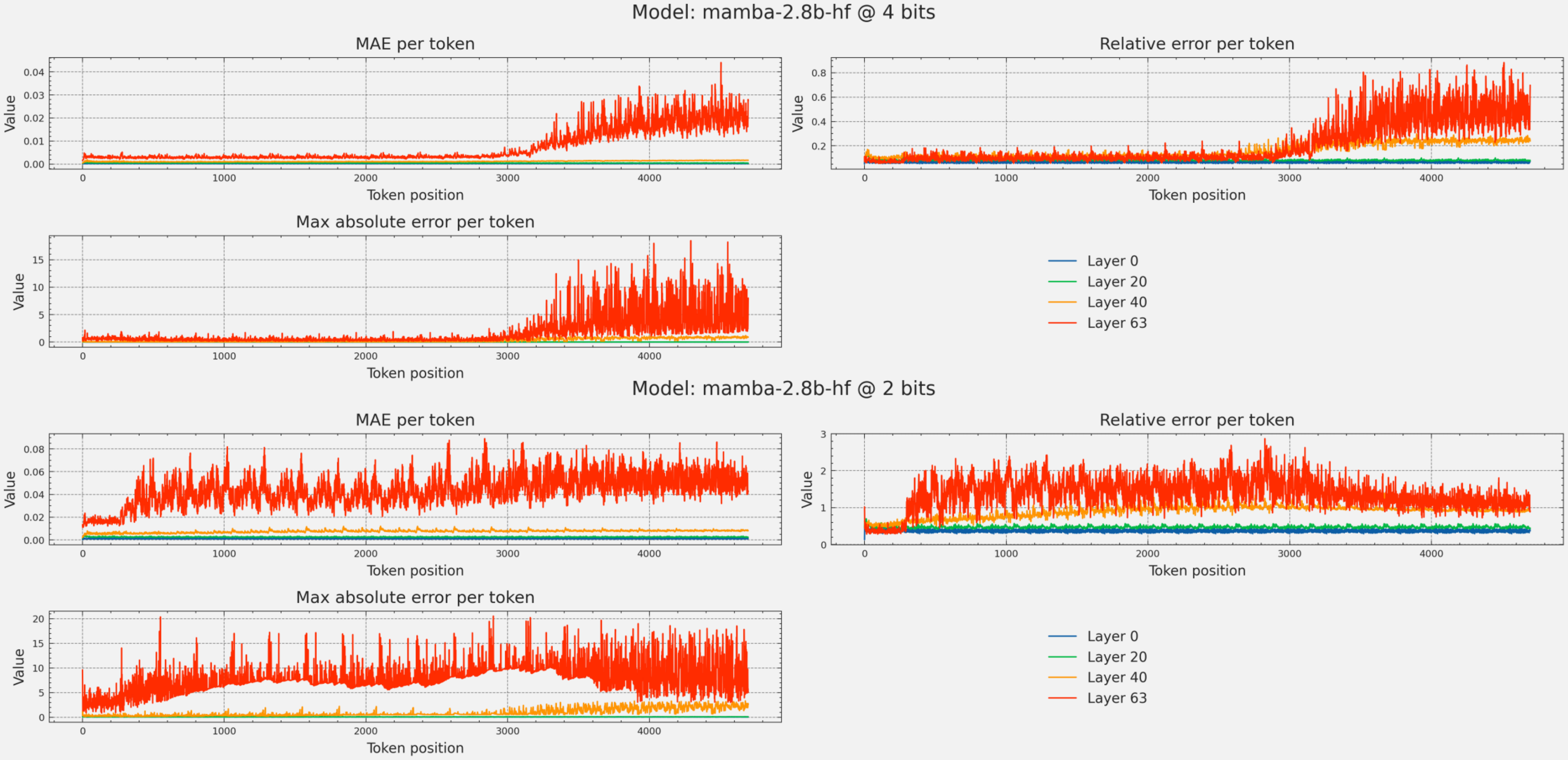
Drift is even more pronounced

Mamba-2.8 / Seq. 4698 custom – rep.

- *Input*: sequence of 261 tokens repeated 18 times
- *Model*: mamba-2.8b-hf
  Considered the state matrix of the SSM block of 4 layers

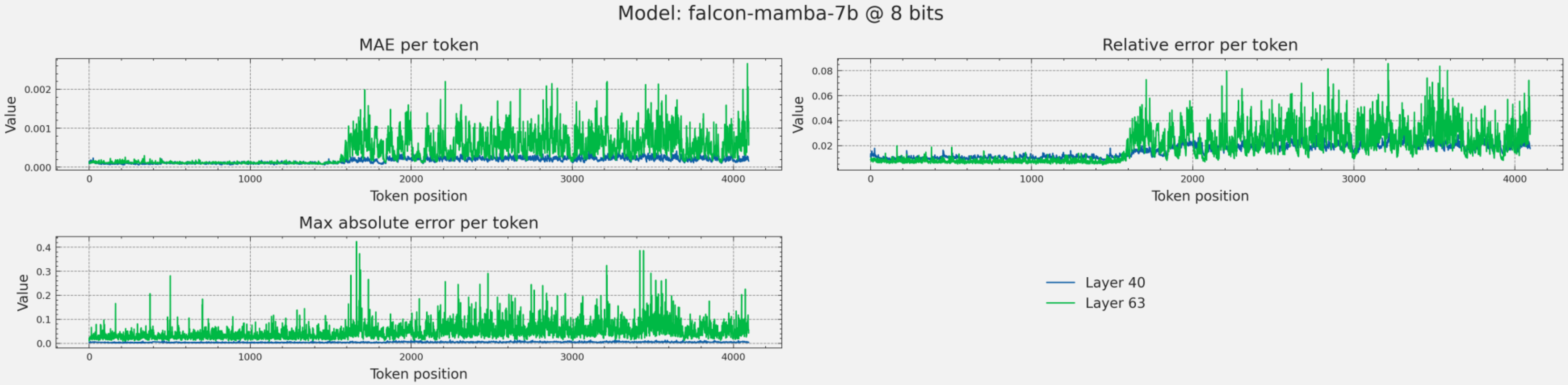


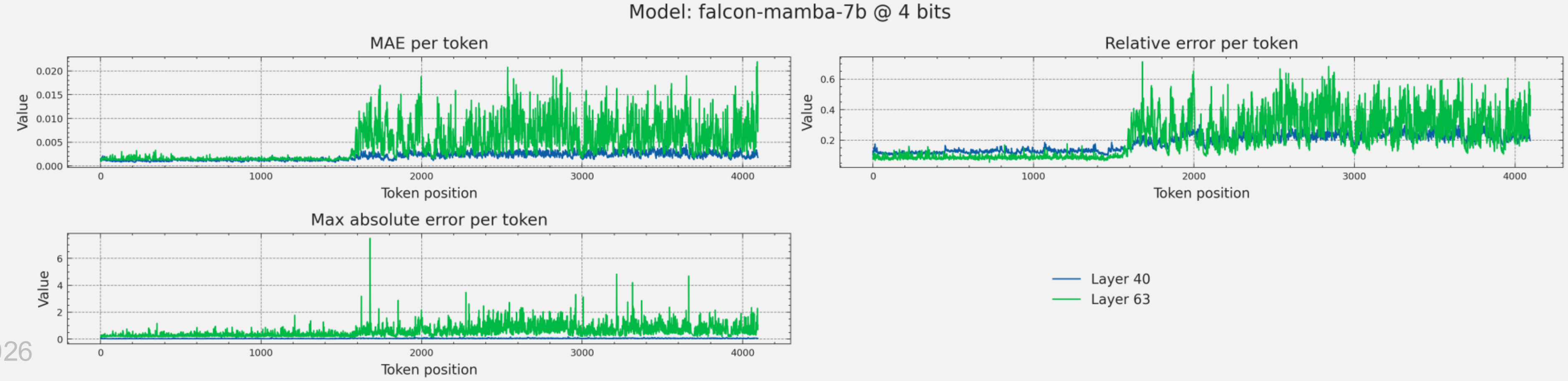
State drift



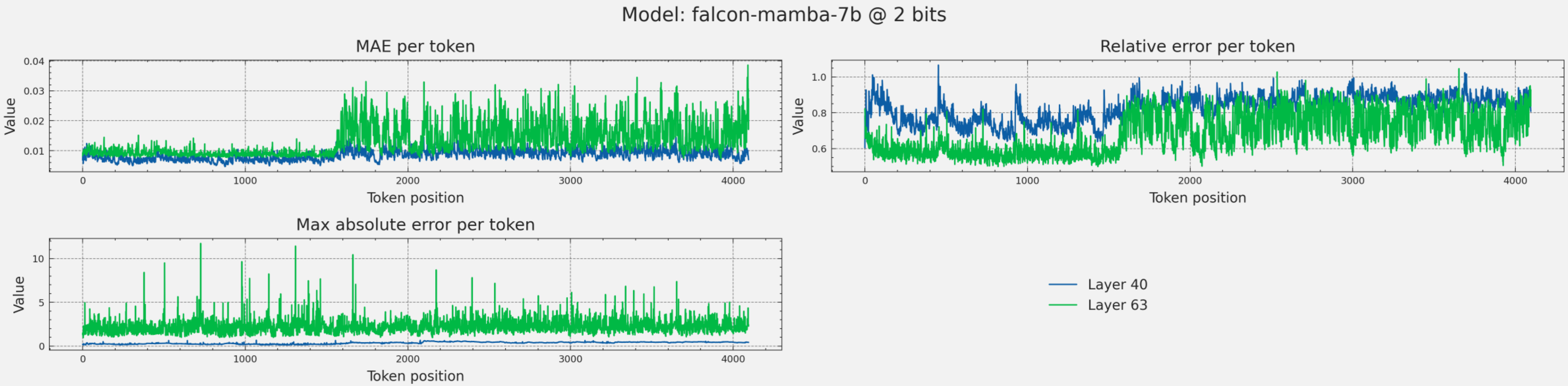
- Last block seems to be the most prone to drifting
- Recurrence is more visible at 2 bits

Falcon-Mamba-7b / Seq. 2048 from Wikitext2





State drift



- TODO:
  - Check correlation with output performance
  - Check with quantized activations

Preliminary results

- Still working on it.
- For now, implemented input activations to the SINQLinear layers quantization
  - Wrapper around SINQLinear to apply RTN (symmetric for now) to the inputs of the layer
  - **SSM block runs in this way in FP (*)** → in TODO list

On Wikitext2

Model	Quamba2			SINQ			
	FP16	W4A8		W4A8 (*)		W4A16	
	PPL	PPL	KLD	PPL	KLD	PPL	KLD
Mamba-2.8b	9.452	10.200	8.084e-2	9.925	5.080e-2	9.714	2.857e-2
Mamba-1.4b	10.747	11.822	9.345e-2	11.281	4.658e-2	11.078	3.086e-2

Next steps

- Continue working on activations quantization
- Hybrid architectures

+ Other paper (WIP)